Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/16/2006
PI Name:	Tarbell, John M. Ph.D.		
Project Title:	Microgravity Effects on Transvascular Transport and Vascular Control		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid transport phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	tarbell@ccny.cuny.edu	**Fax:**	FY 212-650-6727
PI Organization Type:	UNIVERSITY	**Phone:**	212-650-6841
Organization Name:	The City College of New York		
PI Address 1:	Department of Biomedical Engineering		
PI Address 2:			
PI Web Page:	http://bme.engr.ccny.cuny.edu/		
City:	New York	**State:**	NY
Zip Code:	10031	**Congressional District:**	5
Comments:	Former webpage (chgd 4/07): http://fenske.che.psu.edu/		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	03/01/2002	**End Date:**	02/28/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Parsons, Patricia	**Contact Phone:**	216-433-8796
Contact Email:	Patricia.A.Parsons-wingerter@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Tarbell, John. Note coI Norman Harris now has his own grant to finish up his portion of the research.		
COI Name (Institution):	Harris, Norman (Louisiana State University Health Sciences Center)		
Grant/Contract No.:	NAG3-2851		
Performance Goal No.:			
Performance Goal Text:			

When astronauts are exposed to a microgravity environment there is a characteristic tissue fluid shift from the lower extremities to the upper extremities (cephalad fluid shift), which is reversed during re-entry to normal gravity with concomitant reduction in orthostatic tolerance. The biofluid mechanical and biotransport mechanisms which control this fluid shift phenomena are not well understood and will be addressed in the proposed research.
A major fraction of the fluid shift occurs by exchange of water and solutes across the endothelial cell (EC) layer which lines all blood vessels and provides the principal barrier to transport of material between the blood and the underlying tissue. Major alterations in vascular pressure associated with removal (or addition) of the gravitational field changes the transvascular pressure differential, which drives fluid transport between the blood and tissue compartments. We have shown, using a tissue culture model of the EC transport barrier in vitro, that a sudden increase in transvascular water flux induces an early adaptive response in which the hydraulic conductivity (water flux / pressure differential) decreases over a one-hour period. This mechanism tends to maintain constant transmural flow in the presence of changing vascular

Task Description:

pressure conditions. However, these in vitro experiments have further revealed that beyond a one hour exposure to increased pressure, the hydraulic conductivity begins to increase, and reaches a new steady state level which is significantly elevated after about five hours. Furthermore, this increased hydraulic conductivity can be reversed by pharmacological intervention (inhibition of nitric oxide). We hypothesize that this alteration in hydraulic conductivity in response to altered pressure in microgravity contributes to the fluid shift problem. In the proposed research we will study the influence of altered vascular pressure on hydraulic conductivity and albumin permeability of endothelial layers using both in vivo and in vitro models and will explore the biomolecular mechanisms underlying these changes (specific objective 1).

A second important mechanism which controls transvascular fluid exchange is the regulation of blood flow to the tissue experiencing a change in blood pressure. For example, normal control of fluid transport to leg tissue after standing (which increases blood pressure due to the added gravitational force) is associated with the contraction of arterioles in response to increased pressure (the so-called “myogenic response”). The myogenic response limits blood flow and reduces downstream pressure and available surface area, all of which tend to restrict transvascular exchange. This mechanism prevents significant leg edema when we stand. The mechanical factors which drive the myogenic response, however, are not well understood. The traditional view has been that increases in smooth muscle cell (SMC) tension in the blood vessel wall driven by increases in pressure stimulate SMC contraction. However, using an in vitro SMC cell culture model, we have recently demonstrated that fluid flow shear stress on SMC can also drive significant cellular contraction. This strongly suggests that transmural flow, which imposes fluid shear stress on SMC and is driven by changes in vascular pressure as described above, will affect SMC contraction and the myogenic response. We therefore hypothesize that alterations in the myogenic response driven by changes in transvascular flow in altered gravity affect the normal mechanisms which regulate fluid accumulation in tissues. In the proposed research we will study the influence of altered vascular pressure on SMC tone using both in vivo and in vitro models. We will isolate the contributions of vascular tension and fluid flow shear stress to the overall myogenic response and elucidate the underlying biomolecular mechanisms (specific objective 2).

Specific Objective 1. To determine the effects of changes in transvascular pressure on the transport of water and albumin across the endothelial layer. We will use in vivo and in vitro models to approach this specific objective. The in vivo studies will rely upon the microcirculation of the rat mesentery as the experimental model. Chronic pressure changes will be introduced using a micro-perfusion technique and hydraulic conductivity and albumin permeability will be measured over a 6 hour period using the modified Landis technique and fluorescent tracer intensity monitoring. The in vitro studies will employ bovine aortic endothelial cells (BAECs) and bovine retinal microvascular endothelial cells (BRECs) grown on porous supports as models of the endothelial transport barrier. Using a unique apparatus developed in our laboratory, we will monitor the water flux and albumin flux across the endothelial monolayer simultaneously for 6 hours after introducing a change in transendothelial pressure. Changes in the tight junctions and adherens junctions between endothelial cells, which are hypothesized to regulate transendothelial transport, will be monitored using antibodies to the tight junction proteins ZO-1 and occludin and the adherens junction protein, VE-cadherin. Signal transduction pathways involved in remodeling the tight junctions in response to changes in transmural flow (pressure) will then be elucidated.

Specific Objective 2. To determine the effects of changes in transvascular pressure on smooth muscle cell contraction. This specific aim will also be approached using both an in vivo and an in vitro model. The in vivo studies will determine the mechanisms by which myogenic tone is developed in the arterioles of the rat mesentery after sudden (step) and gradual (ramp) changes in pressure. Particular emphasis will be placed on determining the role of transmural interstitial flow in generating myogenic tone. For the first time, we will measure transmural flow during the myogenic response to determine whether it is regulated. In addition we will manipulate transmural flow at a fixed level of pressure change using osmotic agents in order to dissect out, for the first time, the contribution of transmural flow to the development of myogenic tone. The in vitro experiments will employ rat aortic SMC in a 3-dimensional collagen gel model through which controlled interstitial flow will be perfused. Contraction will be observed and quantified under a microscope using video microscopy and image processing, and the biochemical mechanisms mediating the response will be probed with a variety of chemical agonists and inhibitors.

The proposed studies will provide direct understanding of the biotransport and biomolecular mechanisms which control transvascular fluid and solute exchange in response to changes in blood pressure in altered gravity environments. The studies can be extended to humans on the ground or in space and could ultimately lead to effective countermeasures to fluid shifts which impair physiological function in microgravity and upon return to normal gravity.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits: n/a

Specific Objective 1. To determine the effects of changes in transvascular pressure on the transport of water and albumin across the endothelial layer. We used in vivo and in vitro models to approach this specific objective.
The in vitro studies employed bovine aortic endothelial cells (BAECs) and bovine retinal microvascular endothelial cells (BRECs) grown on porous supports as models of the endothelial transport barrier. Using a unique apparatus developed in our laboratory, we monitored the water flux and albumin flux across the endothelial monolayer simultaneously for several hours after introducing a change in transendothelial pressure. Changes in the tight junctions and adherens junctions between endothelial cells, as characterized by the junction protein ZO-1, were monitiored using antibodies. The major finding of the study was that the endothelial transport barrier undergoes a “sealing” effect that lasts on the order of 1 hour after a step increase in pressure is introduced. During the this period both the water flux and solute flux are reduced after an initial step increase associated with the increase in pressure. This appears to be a physiological regulatory effect by which the circulation attempts to protect itself from edema in the face of an increase in pressure. This is important to astronauts in microgravity who experience sudden changes in vascular pressure upon exposure to microgravity or re-entry to normal gravity. A unique finding of our work was that a portion of the sealing effect appears to mediated by purely mechanical factors because they are present even after cells are chemically fixed to eliminate biological activity. On the other hand, a major part of the effect is biologically mediated. We have observed that the junction protein, ZO-1, is actively transported to the junction region from intracellular stores over a time course of 1 hour that corresponds to the time course of the sealing effect. This study provides uniques insights into the mechanisms

Task Progress:	that control fluid shifts in human astronauts exosed to microgravity. The in vivo studies were conducted by Dr. Norman Harris who moved to LSU from Penn State after Dr. Tarbell went to City College from Penn State in 2003. Dr. Harris used the microcirculation of the rat mesentery as the experimental model. Motivated by the in vitro experiments, chronic pressure changes were introduced using a micro-perfusion technique and hydraulic conductivity was measured using the modified Landis technique. A “sealing” effect was also observed in the animal experiments over the same time scale as the in vitro experiments (1 hr). This was a nice in vivo confirmation of the in vitro studies, lending further support to the significance of this work for astronauts. Specific Objective 2. To determine the effects of changes in transvascular pressure on smooth muscle cell contraction. This specific aim was also approached using both an in vivo and an in vitro model. The in vitro experiments employed rat aortic SMC plated on a 2-dimensional surface (glass slide) sub-confluently. Fluid shear stress was applied to the cells in the physiological range and cell contraction was characterized by the change in cell area over time after the imposition of shear stress using video microscopy and image processing. It was observed that significant contraction occurred within 10 min after shear, a time frame consistent with myogenic contractions in arterioles. These contractions also occurred when the shear was imposed in a ramp (over 5 minutes) instead of a step. Mechanism studies showed that the contractions were independent of intracellular calcium, as had been observed in animal studies in vivo. In the animal studies Dr. Harris again used the microcirculation of the rat mesentery and for the first time, measured transmural flow during the myogenic response to determine whether it is regulated. In addition, and most significantly, we manipulated transmural flow at a fixed level of pressure change using osmotic agents in order to dissect out, for the first time, the contribution of transmural flow to the development of myogenic tone. To our great pleasure, we observed that when transmural flow was reduced by 50% by adding osmotic agents, myogenic contractions of arterioles were similarly reduced by 50%. This was the first direct evidence that the myogenic response is controlled by transvascular interstitial flow and not stretch of smooth muscle cells. This has significant implications for the control of vascular exchange and fluid shift during microgravity and re-entry. Task Significance: The studies provided direct understanding of the biotransport and biomolecular mechanisms which control transvascular fluid and solute exchange in response to changes in blood pressure in altered gravity environments. These studies could be extended to humans on the ground or in space and could ultimately lead to effective countermeasures to fluid shifts which impair physiological function in microgravity and upon return to normal gravity.
Bibliography Type:	Description: (Last Updated: 05/18/2006)
Articles in Peer-reviewed Journals	Kim MH, Harris NR, Tarbell JM. "Regulation of hydraulic conductivity in response to sustained changes in pressure." Am J Physiol Heart Circ Physiol. 2005 Dec;289(6):H2551-8. PMID: 16113077 , Dec-2005
Articles in Peer-reviewed Journals	Kim MH, Harris NR, Tarbell JM. "Regulation of capillary hydraulic conductivity in response to an acute change in shear. " Am J Physiol Heart Circ Physiol. 2005 Nov;289(5):H2126-35. PMID: 15994851 , Nov-2005
Articles in Peer-reviewed Journals	Ainslie KM, Garanich JS, Dull RO, Tarbell JM. "Vascular smooth muscle cell glycocalyx influences shear stress-mediated contractile response. " J Appl Physiol. 2005 Jan;98(1):242-9. PMID: 15322072 , Jan-2005
Articles in Peer-reviewed Journals	Ainslie K, Shi ZD, Garanich JS, Tarbell JM. "Rat aortic smooth muscle cells contract in response to serum and its components in a calcium independent manner." Ann Biomed Eng. 2004 Dec;32(12):1667-75. PMID: 15675680 , Dec-2004
Articles in Peer-reviewed Journals	Kim MH, Harris NR, Korzick DH, Tarbell JM. "Control of the arteriolar myogenic response by transvascular fluid filtration." Microvasc Res. 2004 Jul;68(1):30-7. PMID: 15219418 , Jul-2004
Articles in Peer-reviewed Journals	DeMaio L, Tarbell JM, Scaduto RC Jr, Gardner TW, Antonetti DA. "A transmural pressure gradient induces mechanical and biological adaptive responses in endothelial cells." Am J Physiol Heart Circ Physiol. 2004 Feb;286(2):H731-41. PMID: 14527936 , Feb-2004